

Module 3 Lab Reflective Journal

Course: ITAI 1371

Assignment: Module 3 Lab Exercise

John Grubbs

The lab showed me the full start-to-finish process of training a model, at least from the coding perspective. The part I found most useful was learning how data is prepared and divided for classification. The lab separated selected features from the target and then divided the data into 80% training and 20% testing sets. I had not given much thought to the `random_state` parameter before this lab, but I now understand why it is important. By setting a consistent `random_state`, the training and testing data remain the same each time the code is run. This allows different models and approaches to be compared under the same conditions rather than having changes in the data split affect the results.

The course assignments place a lot of emphasis on data collection and cleaning, explaining that these steps can account for around 80% of the work in a machine learning project. However, the datasets in the assignments so far have already been provided and are generally clean, leaving little for us to do beyond executing and modifying the provided code. Because of this, I do not feel that I am getting as much practical experience with these important parts of the process as I would like. I am looking forward to working on a real project where I can collect, organize, categorize, and clean the data myself before applying the modeling techniques introduced in the course.

Luis Adrian

In this lab I learned that supervised learning uses labeled data to make predictions, unsupervised learning looks for patterns in data without known labels, and reinforcement learning improves through actions, rewards, and penalties. These differences helped me see why different machine learning problems will require different approaches. I learned too much work with the Wine dataset, selected features, separated the data into training and testing sets, and trained Logistic Regression and Decision Tree models. In the cells the Logistic Regression achieved 88.9% accuracy, while the Decision Tree achieved 83.3%.

One of the most interesting parts for me was evaluating the models. There are different forms in the classification, but the most interesting one that I saw was "The confusion matrix." and it also helps me to understand which wine classes the model predicted correctly and where it made mistakes. And I also learned that changing the features used to train a model can affect its accuracy, so selecting the right features can be important for improving a better performance.

This lab gave me a better understanding of how the different steps of machine learning connect together. And I would like to learn more about how feature scaling and different classification models can improve results, for the future labs and projects for this class.

Keira Burdine

In this lab, I learned more about the different types of machine learning and how they can be used to solve different problems. The three main types that I worked with were supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labeled data to train a model to make predictions or classifications. Unsupervised learning works with data that does not already have labels and attempts to identify patterns or groups within the data. Reinforcement learning is different because the model learns through experience by receiving feedback or rewards based on its actions. Completing the scenarios in this lab helped me better understand when each type of machine learning would be appropriate.

One of the most helpful parts of the lab was working with the wine classification dataset. I was able to see how different features can affect the performance of a machine learning model. During the experiment, my first selection of features produced an accuracy of 83.3%, which was lower than the original model accuracy of 88.9%. I then experimented with a different combination of features. Using color intensity, OD280/OD315 of diluted wines, and proline increased the model accuracy to 100%. This helped me understand that simply having data is not enough. The features selected to train a model can have a major effect on its performance.

The most challenging part for me was understanding how to determine which features would produce the best results before training the model. I learned through the lab that experimentation and model evaluation are important parts of the machine learning process. Instead of assuming that certain features will work best, the results can be compared using measurements such as accuracy. In the future, I would like to learn more about feature-selection methods that can help identify the most useful features instead of testing different combinations manually.

I was also able to connect machine learning to my interest in cybersecurity. One real-world application I identified was using supervised machine learning to detect potential cyberattacks. A model could be trained using information such as network traffic, login activity, IP addresses, timestamps, and labeled examples of normal and malicious activity. The trained model could then classify new activity and help identify suspicious behavior. This showed me how the concepts I am learning in artificial intelligence can connect directly with cybersecurity.

Overall, this lab gave me a better understanding of the complete machine learning workflow. I learned that building a machine learning model involves more than simply training an algorithm. The process includes defining the problem, collecting and exploring data, preprocessing the data, selecting useful features, training a model, evaluating its performance, deploying it, and continuing to monitor and improve it. The most important concept I learned from this module is how training and testing data work together to build and evaluate a model. As I continue learning machine learning, I would like to become more comfortable with feature selection and understanding why certain models or features perform better than others.

